

ELASTICITY

Elasticity Concepts:

- Meaning of elasticity: If $Y = f(X)$, elasticity measures the *responsiveness* of Y due to changes in X.
- If a given change in X brings about a change in Y, then the elasticity measure attempts to compare the relative change in Y with the relative change in X.
- Mathematical formulation:

$$\varepsilon = \frac{\% \Delta Y}{\% \Delta X} = \frac{\Delta Y / Y}{\Delta X / X}$$

Elasticity Values:

The elasticity values (in absolute terms) can range from zero to infinity; each with definite interpretation:

$ \varepsilon = 0$	Perfectly inelastic
$ \varepsilon < 1$	Inelastic
$ \varepsilon = 1$	Unit elastic
$ \varepsilon > 1$	Elastic
$ \varepsilon = \infty$	Perfectly elastic

Demand Elasticity Concepts:

- We shall study 3 demand elasticity concepts:
 - **Own price elasticity** : responsiveness of quantity demanded of a good to changes in own-price.
 - **Cross price elasticity** – responsiveness of quantity demanded of good A to changes in price of good B (substitute or complement).
 - **Income elasticity** – responsiveness of quantity demanded of a good due to changes in income.

Own-Price Elasticity:

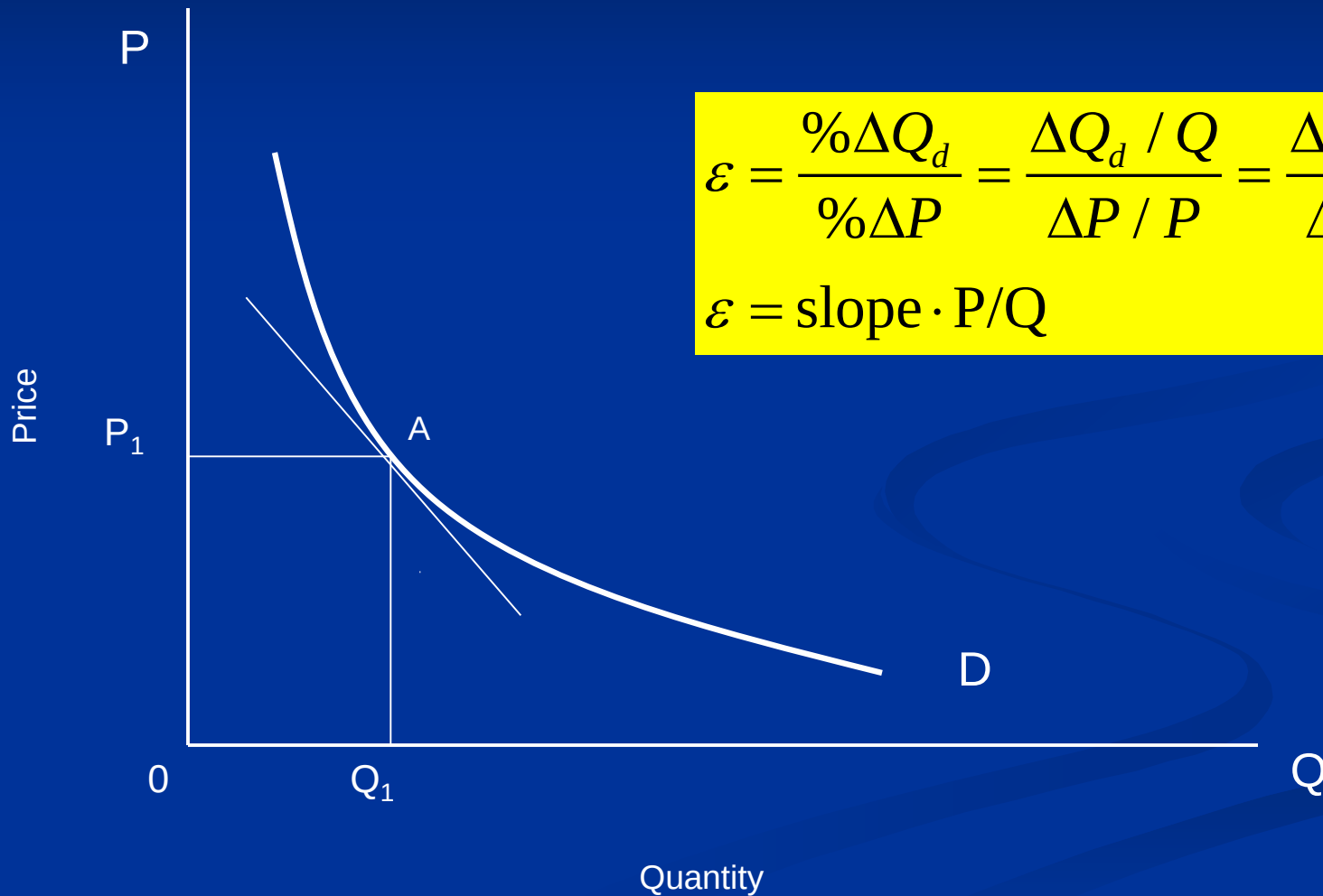
- *Own price elasticity of demand* (ϵ) – measure of responsiveness of quantity demanded to changes in price:

$$\epsilon = \frac{\% \Delta \text{in } Q_d}{\% \Delta \text{in } P}$$

Two ways of measuring elasticity:

- ***Point elasticity*** $\Rightarrow$ elasticity is measured for a single point.
 - *More precise since elasticity changes at every point on demand curve.*
 - *Can be obtained if demand function is known.*
- ***Arc elasticity*** $\Rightarrow$ computed for two points along a demand curve
 - Done if we have limited number of observations.

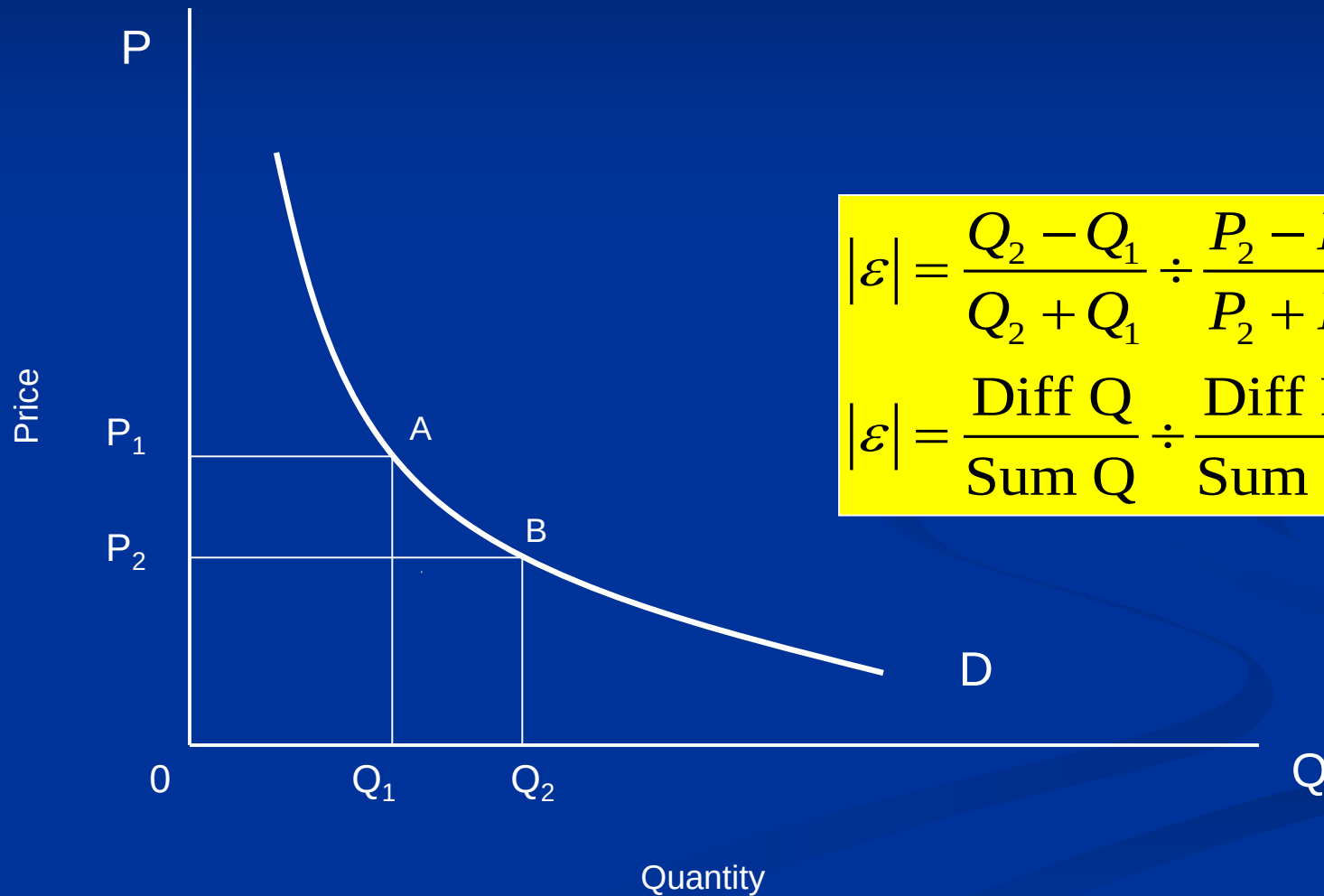
Point Elasticity:



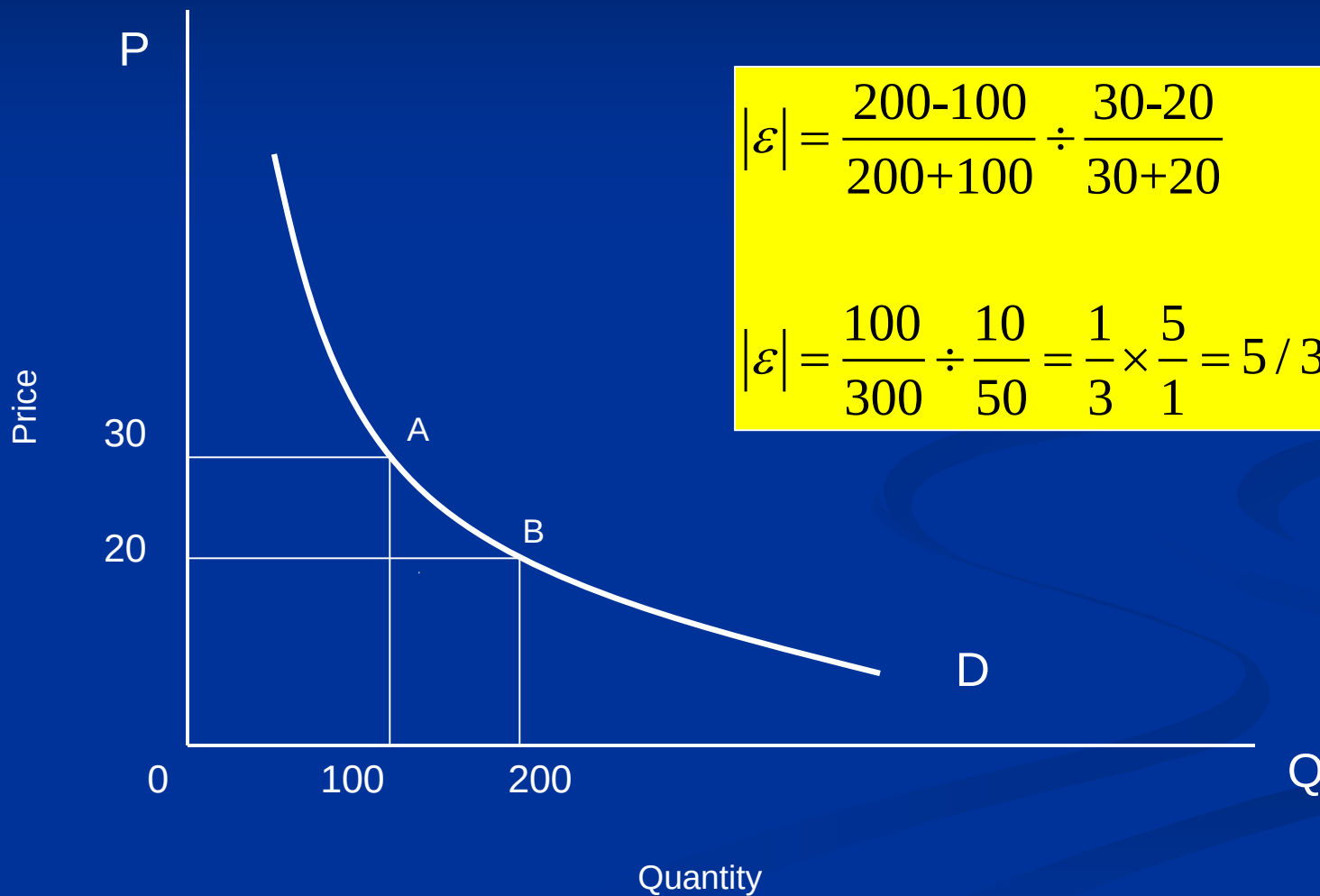
$$\varepsilon = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{\Delta Q_d / Q}{\Delta P / P} = \frac{\Delta Q_d}{\Delta P} \cdot \frac{P}{Q}$$

$$\varepsilon = \text{slope} \cdot P/Q$$

Arc Elasticity:

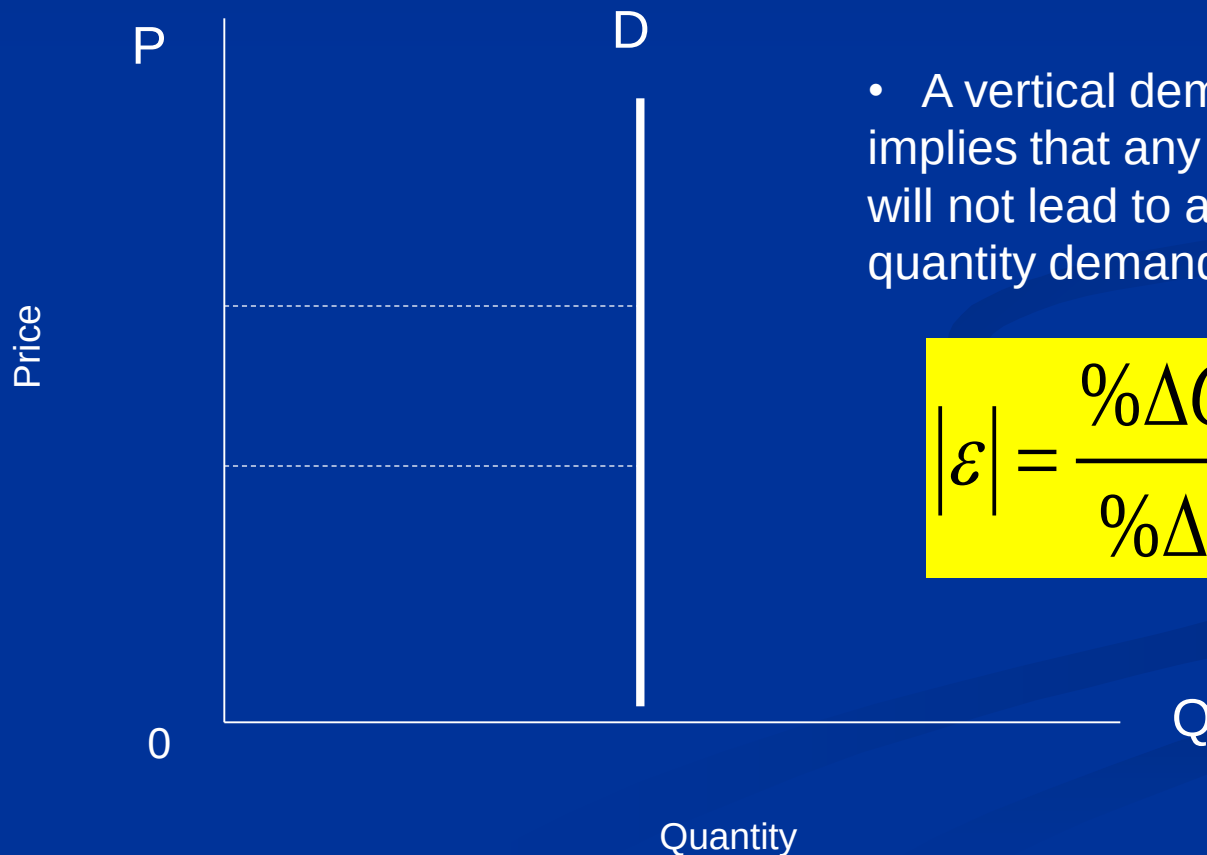


Arc Elasticity: Example



Special Case 1:

■ Perfectly Inelastic Demand Curve:

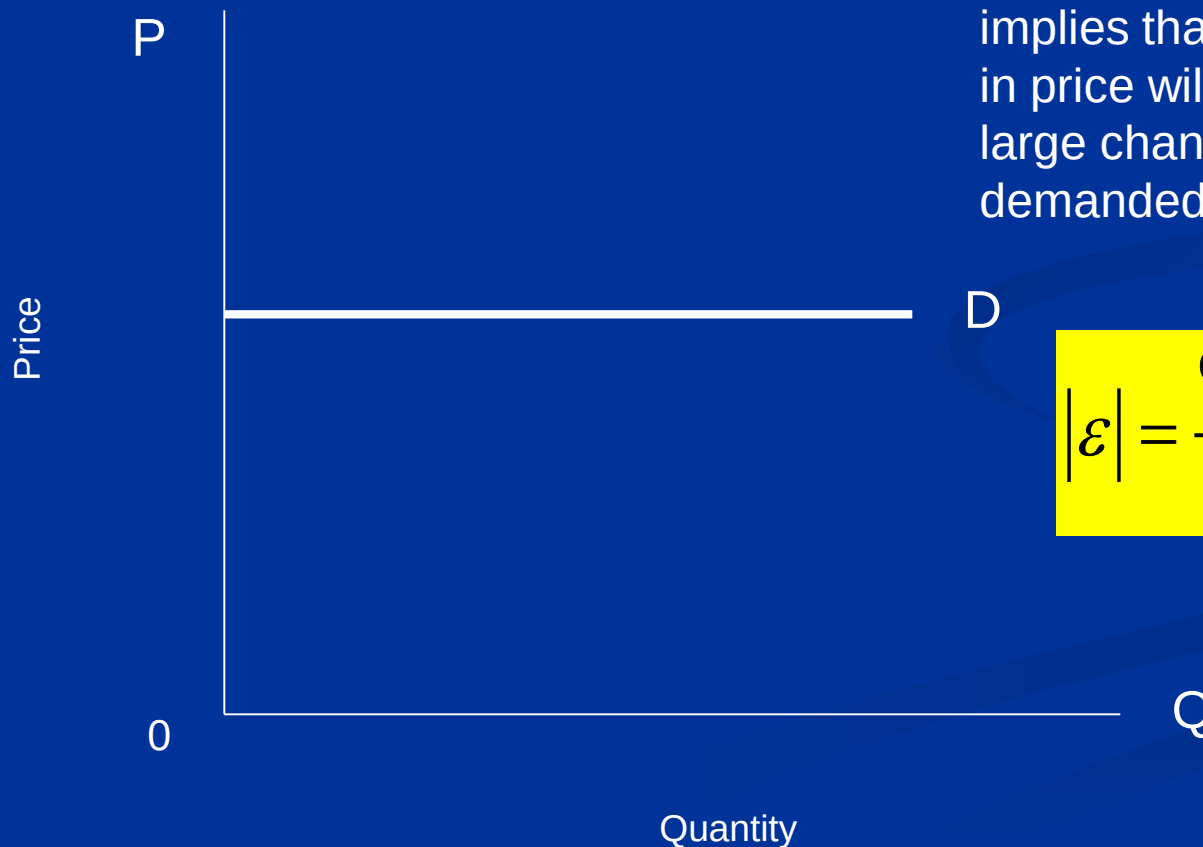


- A vertical demand curve implies that any change in price will not lead to a change in quantity demanded.

$$|\epsilon| = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{0}{+} = 0$$

Special Case 2:

■ Perfectly Elastic Demand Curve:



- A horizontal demand curve implies that a very small change in price will lead to an infinitely large change in quantity demanded.

$$|\epsilon| = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{+}{0} = \infty$$

Elasticity and Total Revenue:

- Total revenue (from the point of view of a seller) is equal to the quantity sold multiplied by its price.
- $TR = P \times Q$
- It is of interest to the seller what happens to TR if price is increased or decreased, knowing that if it happens, consumers will adjust their purchases.

What happens to TR when price increases?

Answer: it depends on the elasticity of demand

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta P}$$

Elastic	P ↑ Q ↓	TR decreases
Unitary	P ↑ Q ↓	TR unchanged
Inelastic	P ↑ Q ↓	TR increases

What happens to TR when price decreases?

Answer: it depends on the elasticity of demand

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta P}$$

Elastic	P ↓ Q ↑	TR increases
Unitary	P ↓ Q ↑	TR unchanged
Inelastic	P ↓ Q ↑	TR decreases

Determinants of Price Elasticity of Demand:

- (1) the availability of good substitutes for the commodity
 - $\Rightarrow$ more substitutes, more elastic.
- (2) the number of uses the good can be put into
 - $\Rightarrow$ more uses, more elastic.
- (3) the price of the good relative to the consumer's income
 - $\Rightarrow$ if a good takes a larger share of budget, likely to be more elastic.
- (4) the time frame under consideration
 - $\Rightarrow$ longer period of time, more elastic (Second Law of Demand).

Cross-Price Elasticity of Demand:

- Definition: responsiveness of quantity demanded of a good to changes in price of other goods.
- Formula:

$$e_{xy} = \frac{\% \Delta Q_{dx}}{\% \Delta P_y}$$

- Sign: + for substitutes
- for complements
0 for independent (unrelated) goods

Income Elasticity of Demand:

- Definition: responsiveness of quantity demanded of a good to changes in income.
- Formula:

$$\eta_{xI} = \frac{\% \Delta Q_{dx}}{\% \Delta I}$$

- Sign: + for normal (superior) goods
- for inferior goods

Some Applications of Supply and Demand and Elasticity:

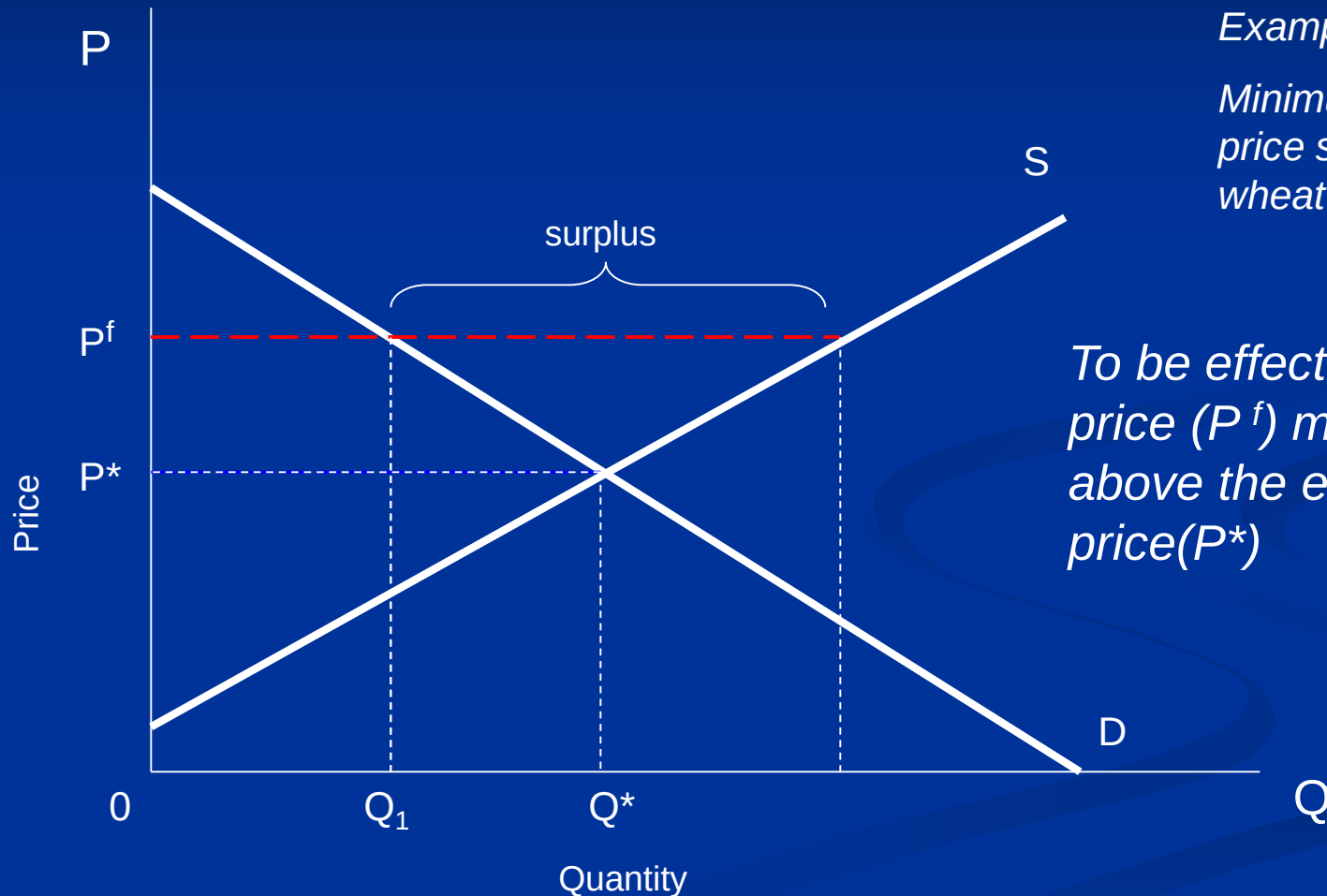
- **Minimum Price Policy** – floor prices to protect producers (price support) or workers (minimum wage).
- **Maximum Price Policy** – price ceilings to protect consumers (rent, electricity, fuel, ... etc.).
- **Tax Incidence** – who bears the burden when tax is imposed on the producer?

Minimum (Floor) Price Policies:

- Prices cannot go below a specified level, e.g. price support for agricultural commodities, minimum wages ...etc.
- Floor price is usually set above the equilibrium price and it causes a surplus.

Floor Price

(minimum price policy)



Examples:

*Minimum wages,
price support for
wheat farmers*

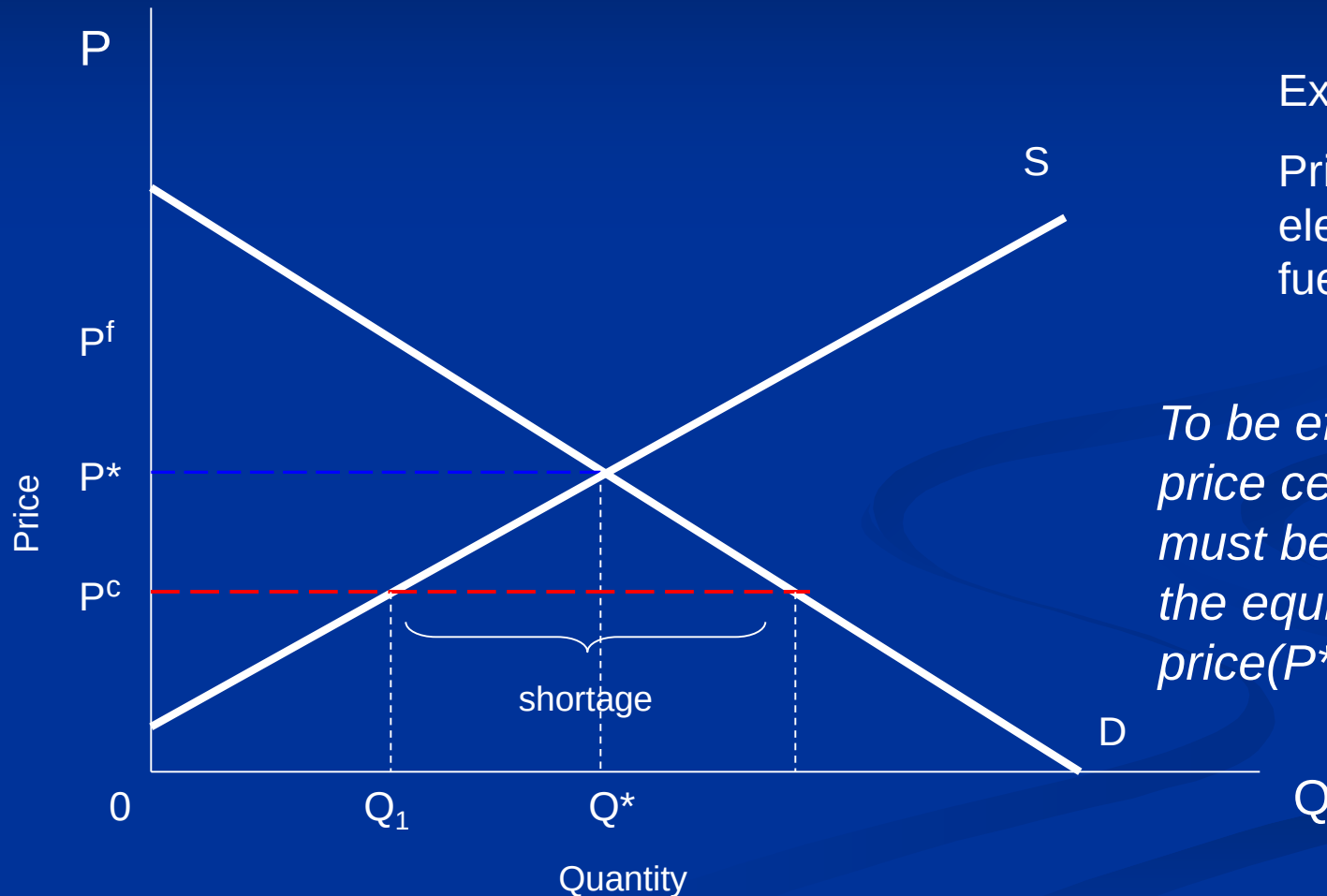
*To be effective, a floor
price (P^f) must be set
above the equilibrium
price (P^*)*

Maximum (Ceiling) Price Policy :

- Price cannot be charged above a specified level.
- Example: maximum rent, maximum transport fare, etc.
- Usually set below the equilibrium price and causes a shortage.

Price Ceiling

(maximum price policy)



Examples:

Price control of
electricity, rents,
fuel

*To be effective, a
price ceiling (P^c)
must be set below
the equilibrium
price (P^*)*

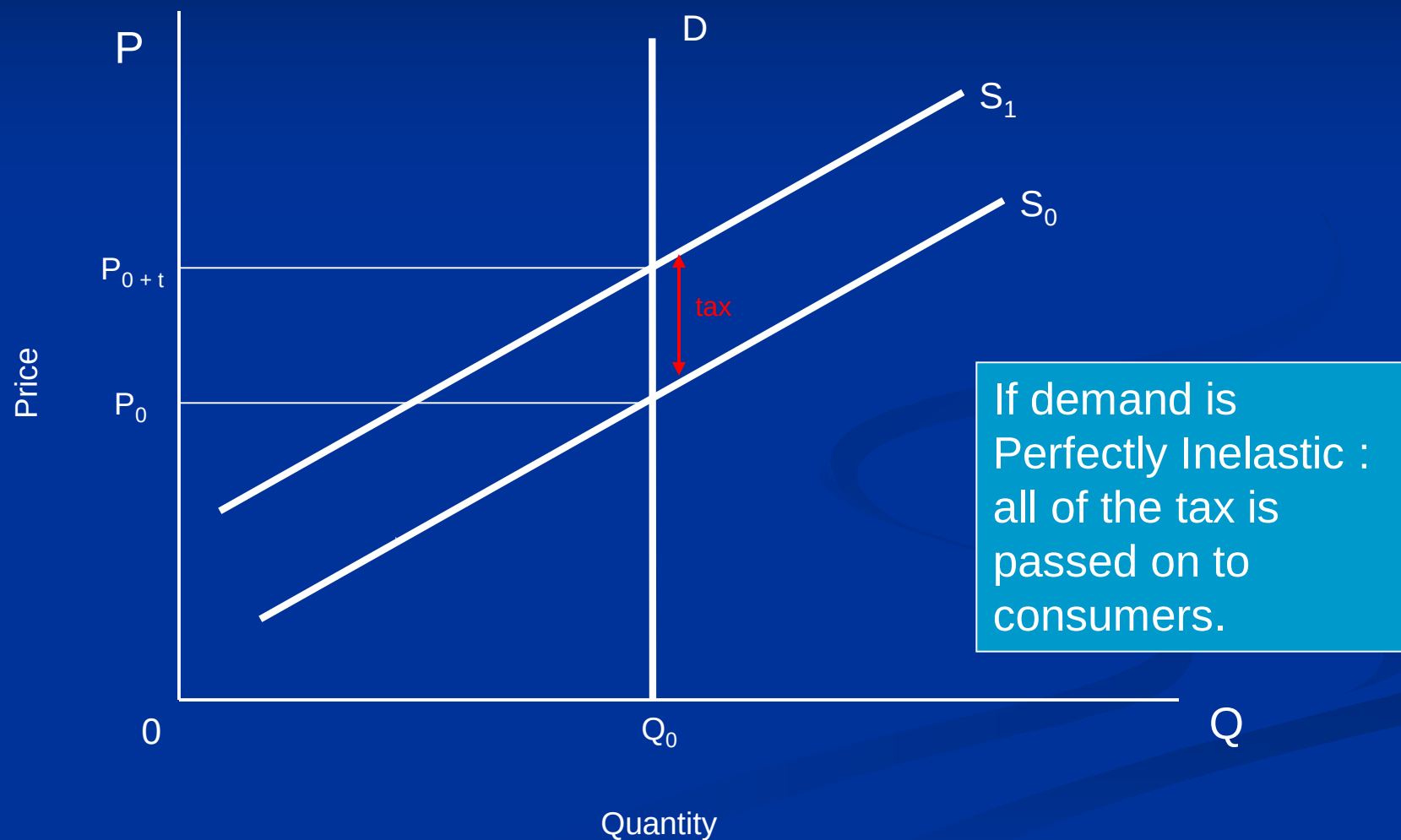
Tax incidence:

- Concerned with effects of government tax policies on consumption and production.
- The tax could either be a *specific (excise) tax* or an *ad valorem tax*.
 - specific tax or excise tax $\Rightarrow$ tax per unit of the product (e.g. SDG 100 per ton).
 - ad valorem tax $\Rightarrow$ tax as percentage of the selling price (e.g. 10% of price).

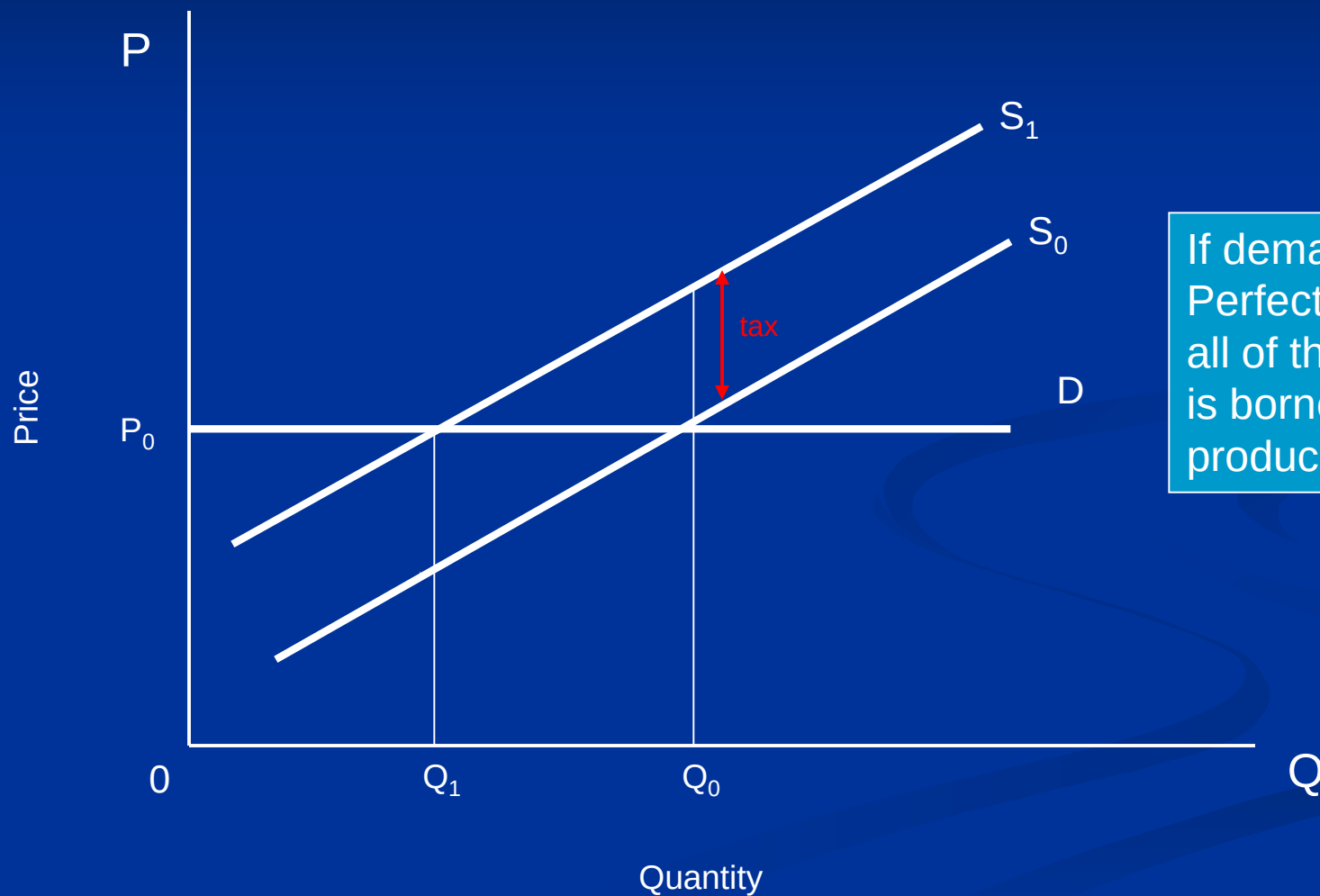
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- Question: Who bears the greater portion of tax? Is it the consumer or the producer?
 - The tax is likely to be paid for by both producers and consumers.
 - The tax is likely to raise the equilibrium price, but by an amount less than the tax.

Tax Incidence: Perfectly Inelastic Demand:

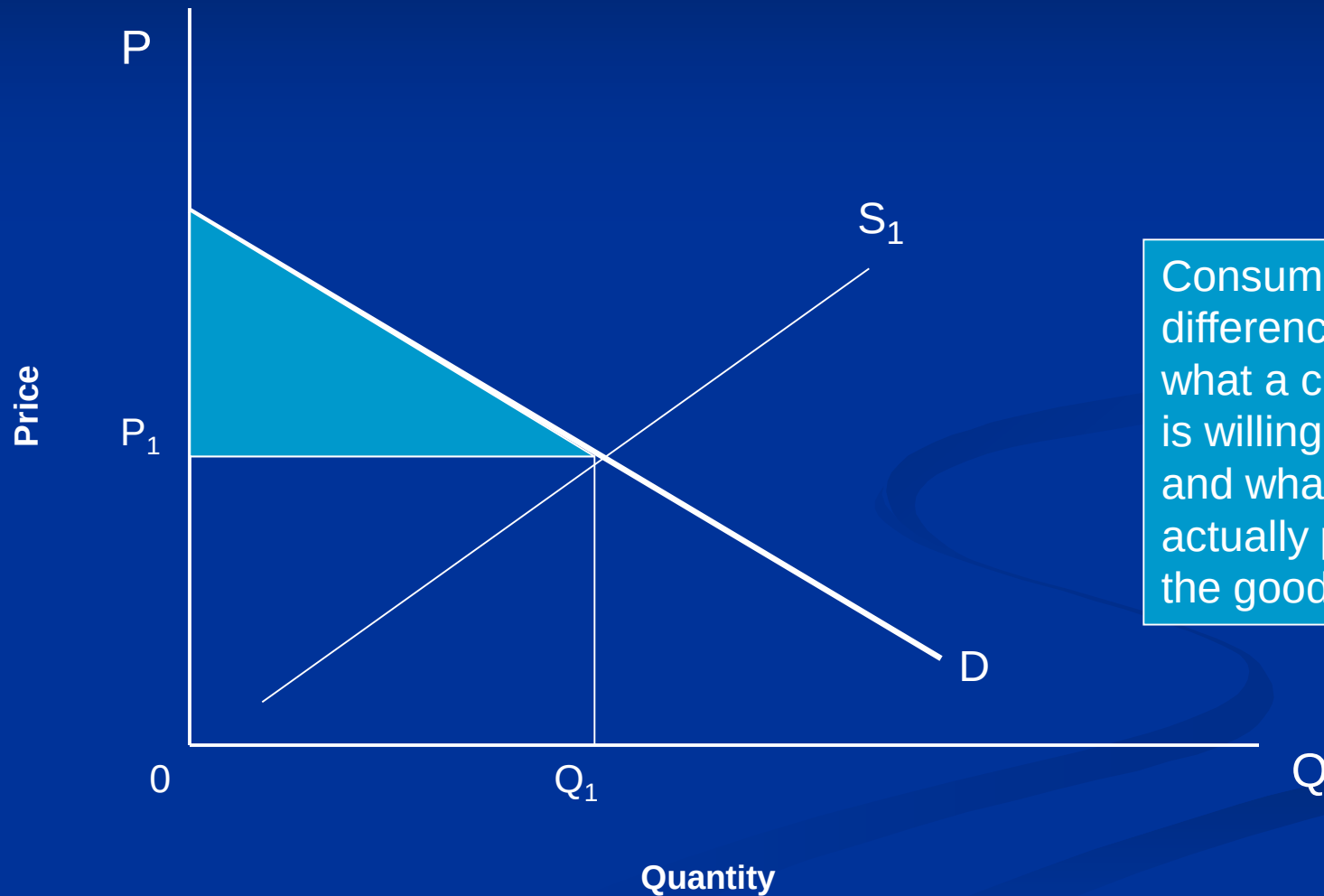


Tax Incidence: Perfectly Elastic Demand



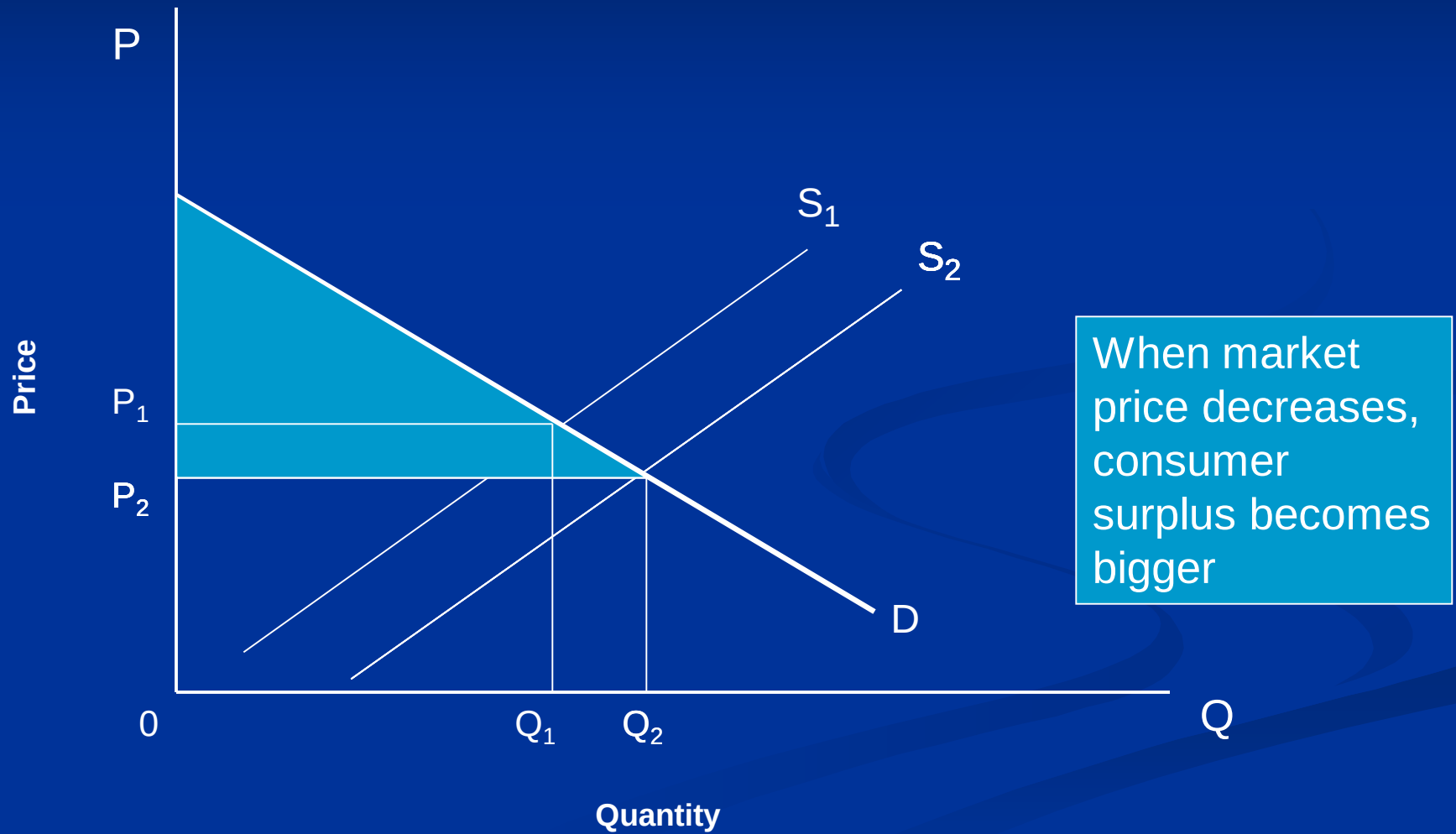
If demand is Perfectly Elastic : all of the tax burden is borne by the producer.

Consumer Surplus:

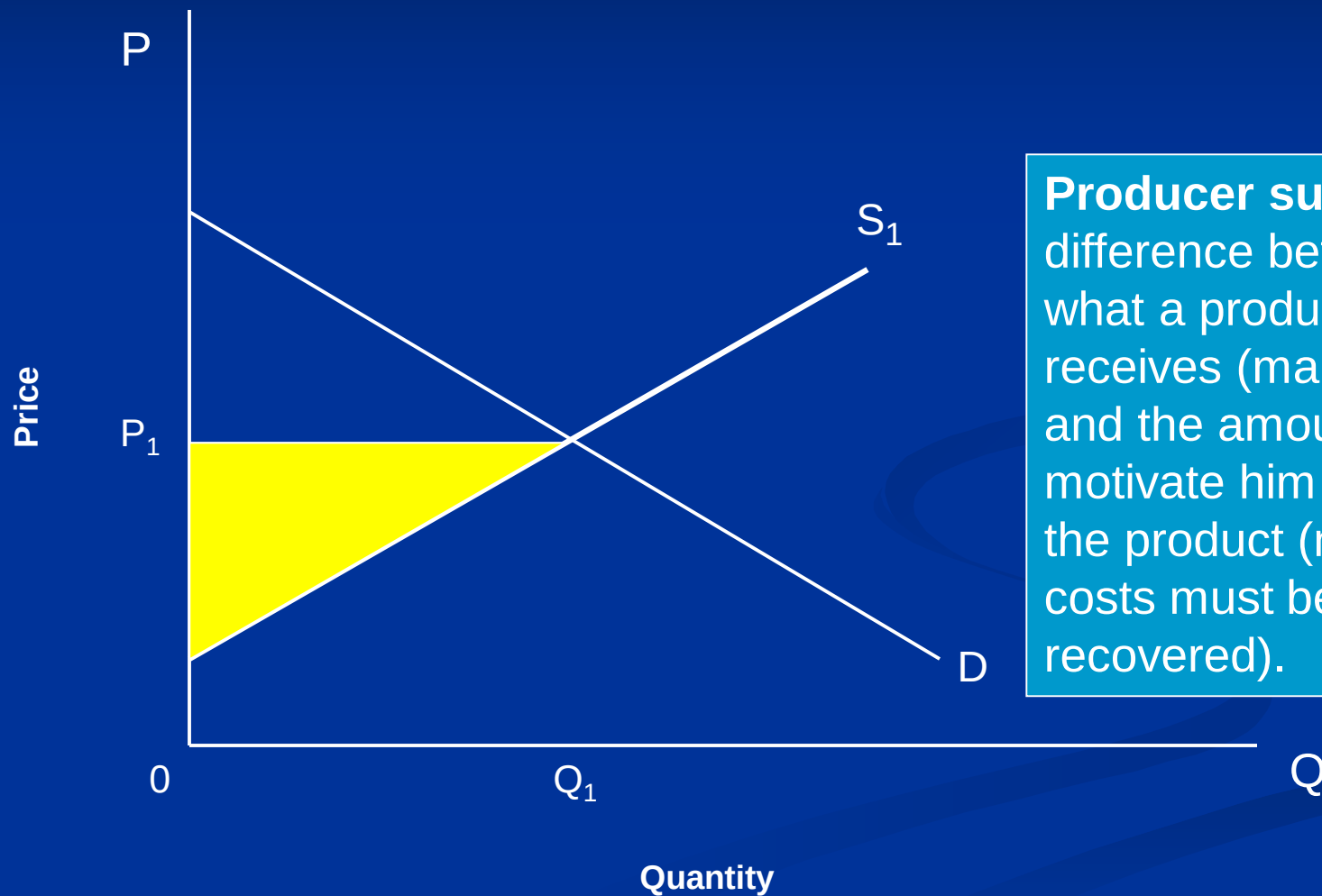


Consumer surplus: difference between what a consumer is willing to pay and what he actually pays for the good.

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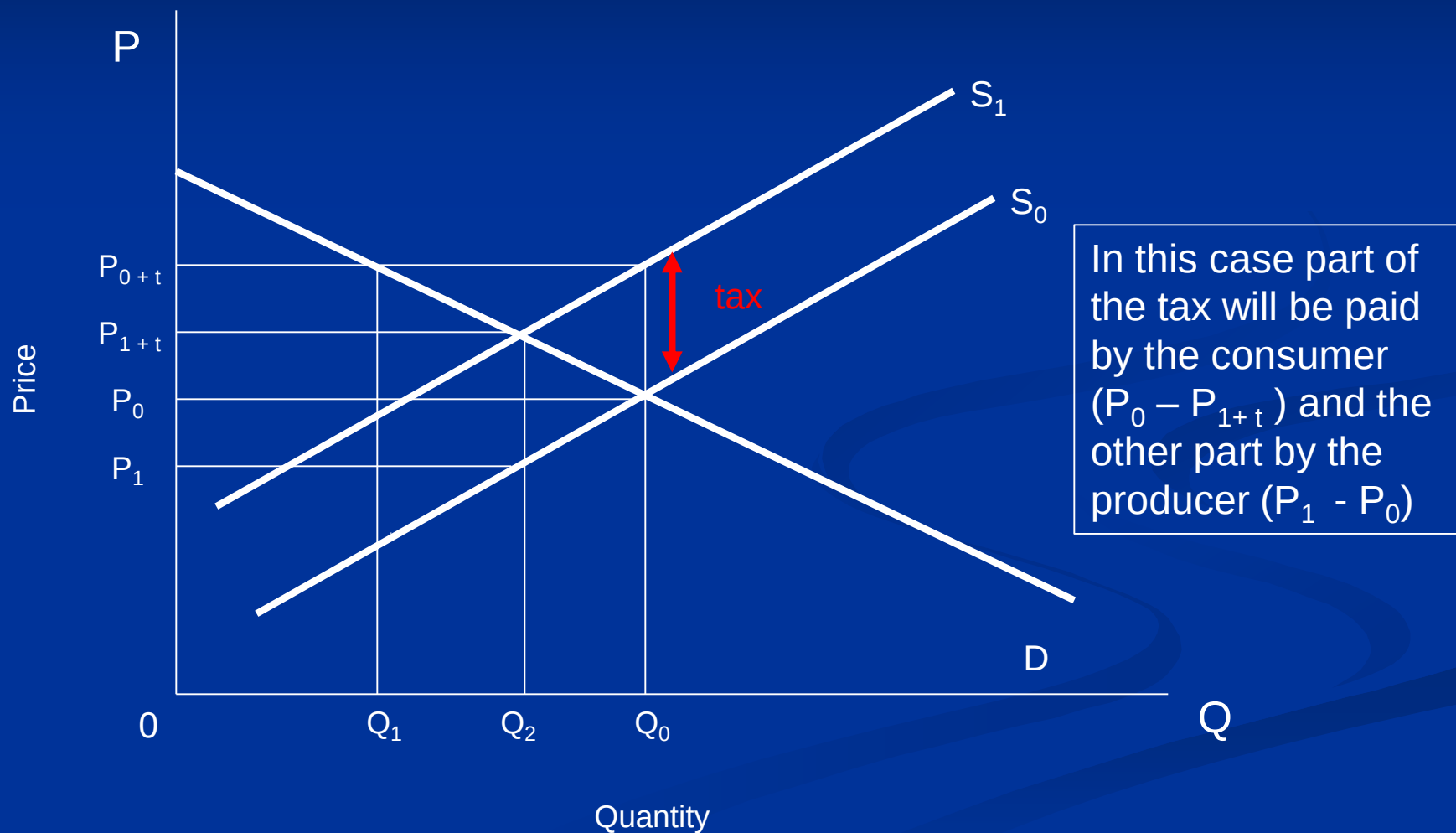


Producer Surplus:



Producer surplus: difference between what a producer receives (market price) and the amount that will motivate him to supply the product (marginal costs must be recovered).

Tax Incidence – Elastic Demand and Supply:



Readings: Samuelson & Nordhaus, Ch. 4